

Assignment 3: Classification using the Naïve Bayes Algorithm

Dataset: Student Placement Dataset
Programming Language: Python
Level: M.Tech. 1st Year

Objective: To understand and implement the Naïve Bayes classification algorithm from scratch in Python by estimating class priors, feature-wise likelihoods, and posterior scores, and to evaluate the classifier on a student placement dataset.

Task Instructions:

❖ Dataset Loading and Preprocessing

- Load **Assignment_3_Naive_Bayes_Dataset.csv** using Python.
- Identify the target variable (**Placed**) and separate it from the input features.
- Check the dataset for missing values. Handle missing numerical values using an appropriate method such as mean imputation. Clearly state the method used.
- Do not use a ready-made Naïve Bayes classifier for the main implementation.

❖ Train-Test Split

- Split the dataset into training and testing sets using an **80:20** split.
- Use a fixed random seed so that the experiment is reproducible.
- Report the number of training and testing samples and the class distribution in both sets.

❖ Implement Naïve Bayes from Scratch

- Implement a Python class or equivalent functions for a **Gaussian Naïve Bayes** classifier using NumPy/Python.
- Calculate the prior probability $P(C)$ for each class.
- For every numerical feature and every class, calculate the **mean** (μ) and **variance** (σ^2) from the training data.
- Implement the Gaussian probability density function to calculate the likelihood $P(x_i|C)$.
- For a test instance, calculate the class score using the prior and the product of the feature likelihoods.
- Use logarithms of probabilities when appropriate to reduce numerical underflow, and select the class with the highest posterior score.
- Implement **fit()** and **predict()** functionality (or an equivalent structure) yourself.

❖ Testing and Evaluation

- Use the trained classifier to predict the class labels for the test set.
- Calculate and report **accuracy, precision, recall, and F1-score**.
- Display the confusion matrix and interpret TP, TN, FP, and FN for the **Placed = Yes** class.
- Report at least five test instances with their actual class, predicted class, and posterior/log-posterior scores for the two classes.

❖ Verification and Analysis

- Manually verify the calculation for **one test instance**: show its prior, individual feature likelihoods, class score, and final predicted class.
- Compare your scratch implementation with **sklearn.naive_bayes.GaussianNB** only as an optional verification step. Explain any difference in predictions or metrics.

- Remove one feature at a time and observe how the test accuracy changes. Identify which feature appears most useful for classification.
- Discuss the conditional-independence assumption of Naïve Bayes and whether it is realistic for student placement data.
- Explain why the algorithm is called “Naïve” and mention at least two advantages and two limitations.

❖ **Analysis and Conclusion**

- Summarize the obtained performance metrics and major observations.
- Comment on the effect of missing-value preprocessing and the 80:20 split.
- Conclude whether Naïve Bayes is suitable for this dataset and justify your answer using your experimental results.

Questions to Answer in the Submission:

1. What are prior probability, likelihood, and posterior probability in Naïve Bayes?
2. Write the Naïve Bayes classification rule and explain the conditional-independence assumption.
3. Why is a Gaussian distribution appropriate for the continuous features in this assignment?
4. Why should multiplication of many small probabilities be handled carefully in Python? Explain the use of log-probabilities.
5. What would happen if a variance becomes zero for a feature within a class? How would you handle it safely?
6. Explain the meaning of the confusion matrix obtained for the placement prediction problem.
7. Which feature contributed most to the classifier according to your feature-removal experiment? Give evidence from your results.

Submission Requirements:

- Python source code (.py or Jupyter Notebook).
- The completed CSV dataset supplied with this assignment.
- Output showing preprocessing, learned parameters, predictions, confusion matrix, and evaluation metrics.
- A brief discussion/conclusion (approximately 1–2 pages).

Important: The supplied dataset is an educational dataset created for this assignment. Students must implement the Naïve Bayes algorithm themselves; using a library's ready-made classifier is permitted only for optional verification/comparison.